

Plots and diagnostic outputs used for the missingness analysis and validation.

Task 1

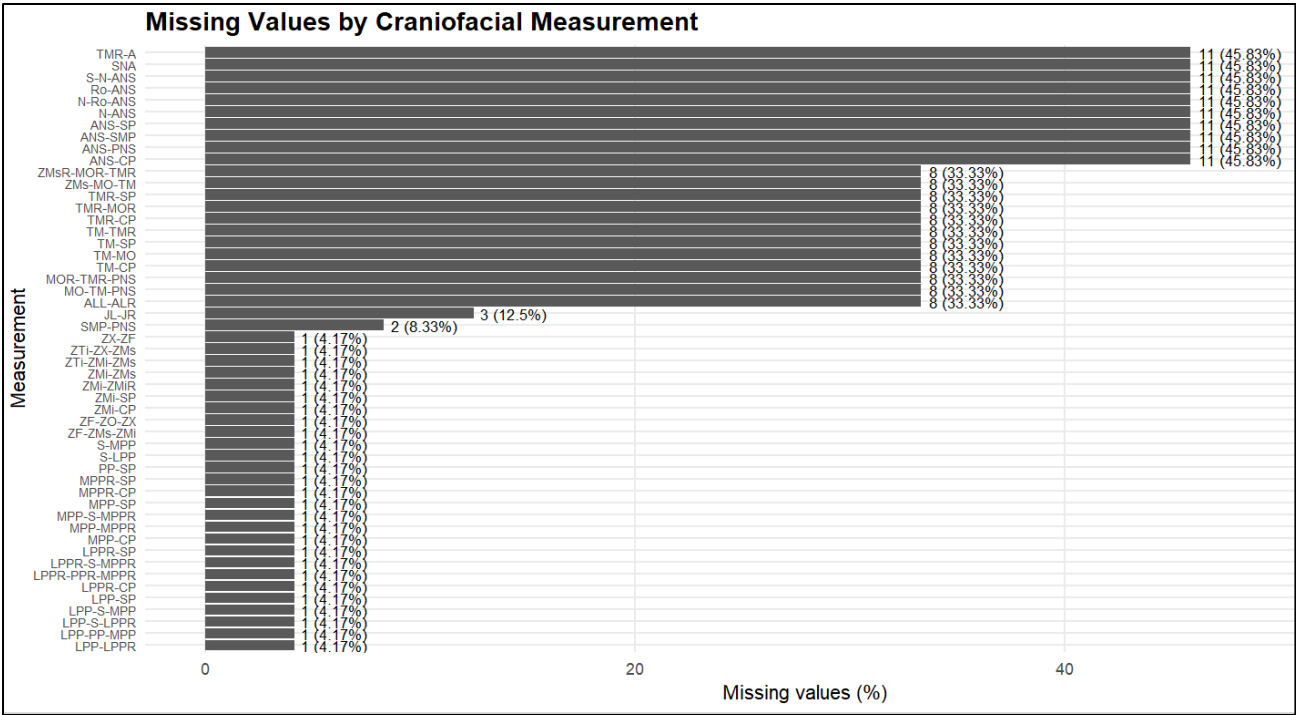


Figure 1. Percentage of missing values by craniofacial measurement. Only measurements with at least one missing observation are shown; labels indicate the number and percentage of missing observations.

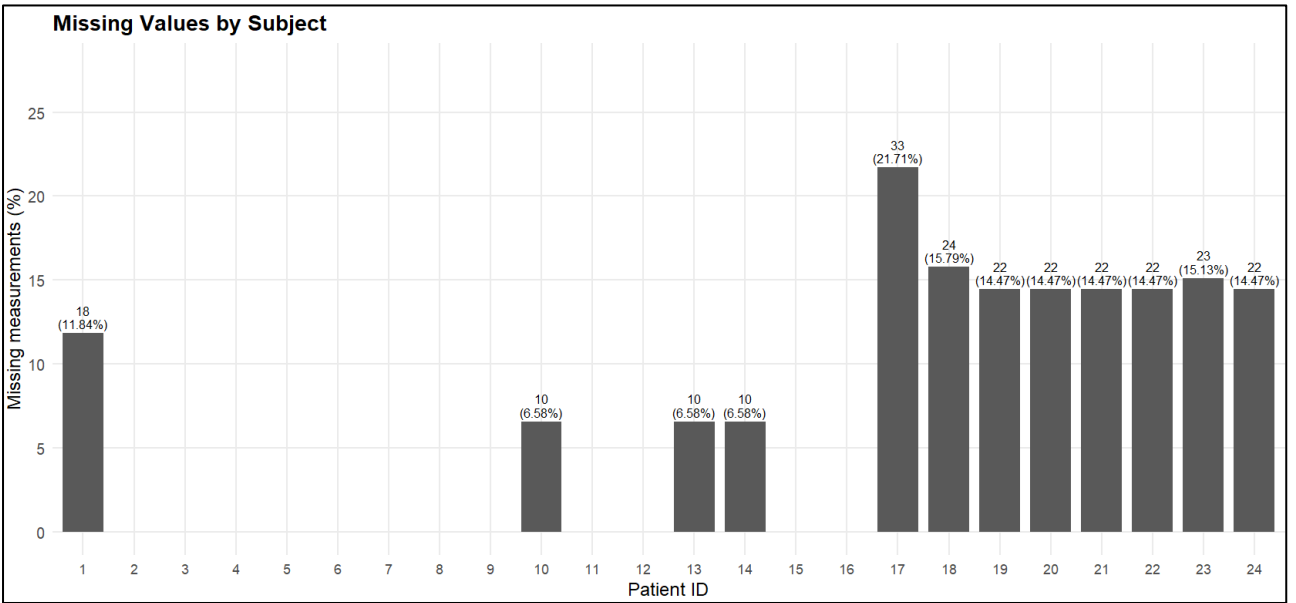


Figure 2. Percentage of missing craniofacial measurements by subject. Labels indicate the number and percentage of missing measurements for each subject.

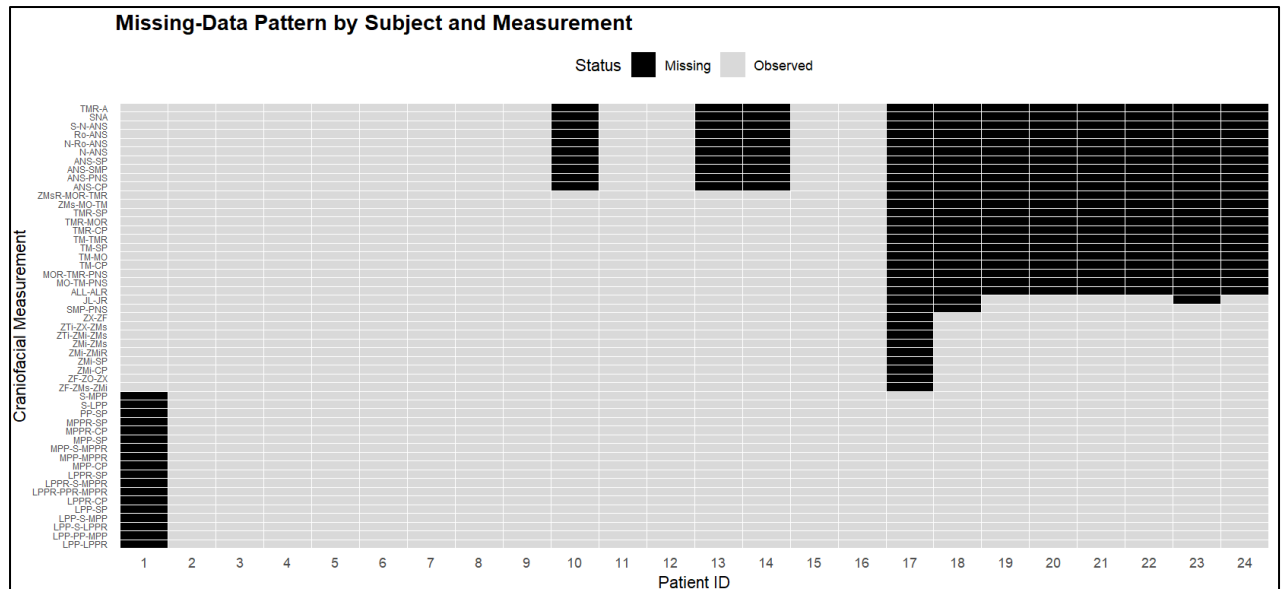


Figure 3. Missing-data pattern by subject and craniofacial measurement. Black cells indicate missing observations and grey cells indicate observed observations. Only measurements with at least one missing value are displayed.

Table 1. Shared Missingness Patterns Among Craniofacial Measurements and Measurement Pairs

Relationship examined	Measurements	Missing patients
Shared ANS label	ANS-CP, ANS-PNS, ANS-SMP, ANS-SP, N-ANS, N-Ro-ANS, Ro-ANS, S-N-ANS	10, 13, 14, 17–24
Shared LPP/LPPR/MPP/MPPR labels	LPP/LPPR/MPP/MPPR-related measurements	Patient 1
Apparent left/right pair	TM-CP / TMR-CP	17–24 / 17–24
Apparent left/right pair	TM-SP / TMR-SP	17–24 / 17–24
Apparent left/right pair	TM-MO / TMR-MOR	17–24 / 17–24
Apparent left/right pair	MO-TM-PNS / MOR-TMR-PNS	17–24 / 17–24
Apparent left/right pair	ZMs-MO-TM / ZMsR-MOR-TMR	17–24 / 17–24
Apparent left/right pair	ZMi-CP / ZMiR-CP	17 / none
Apparent left/right pair	ZMi-SP / ZMiR-SP	17 / none

Task 2

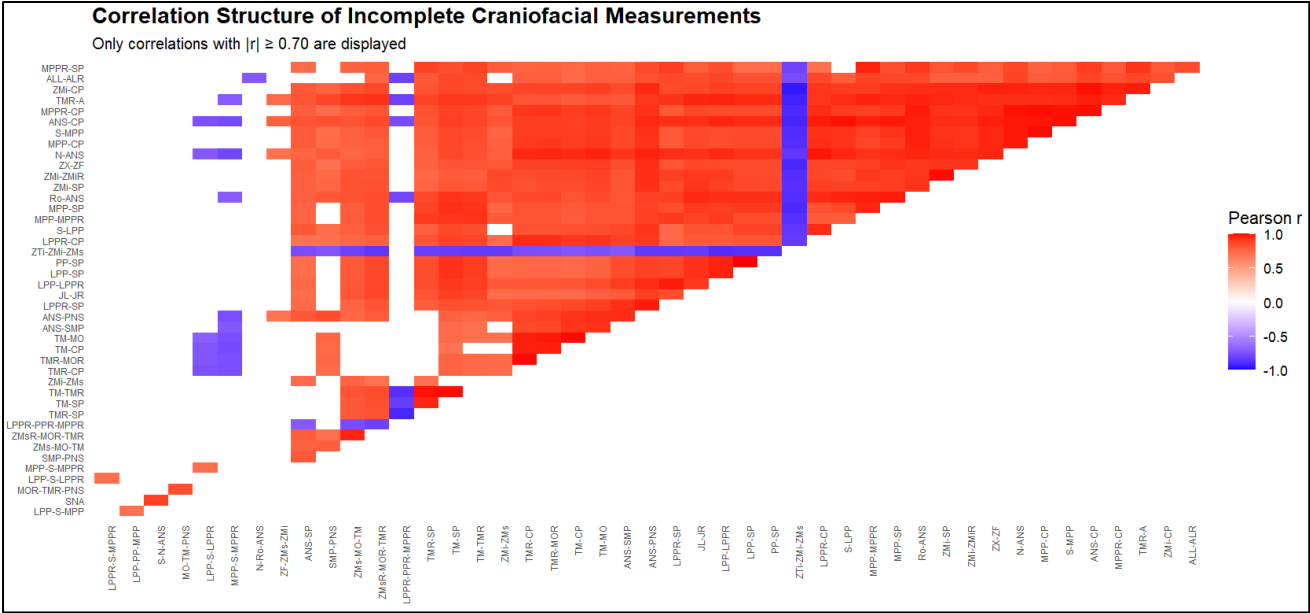


Figure 4. Correlation structure among craniofacial measurements containing missing values. Only Pearson $|r| \geq 0.70$ correlations with are displayed.

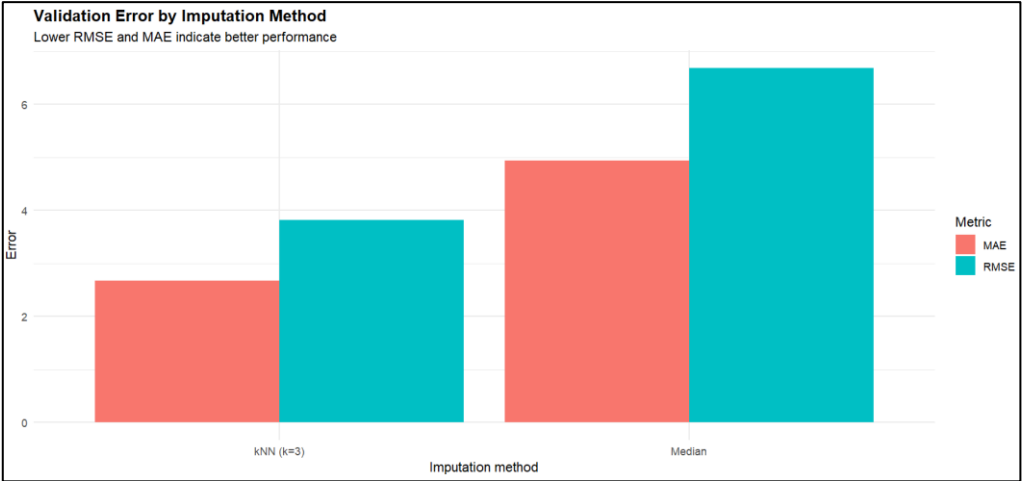


Figure 5. Validation performance of median and kNN imputation based on artificially masked observed values. Lower RMSE and MAE indicate better predictive performance.

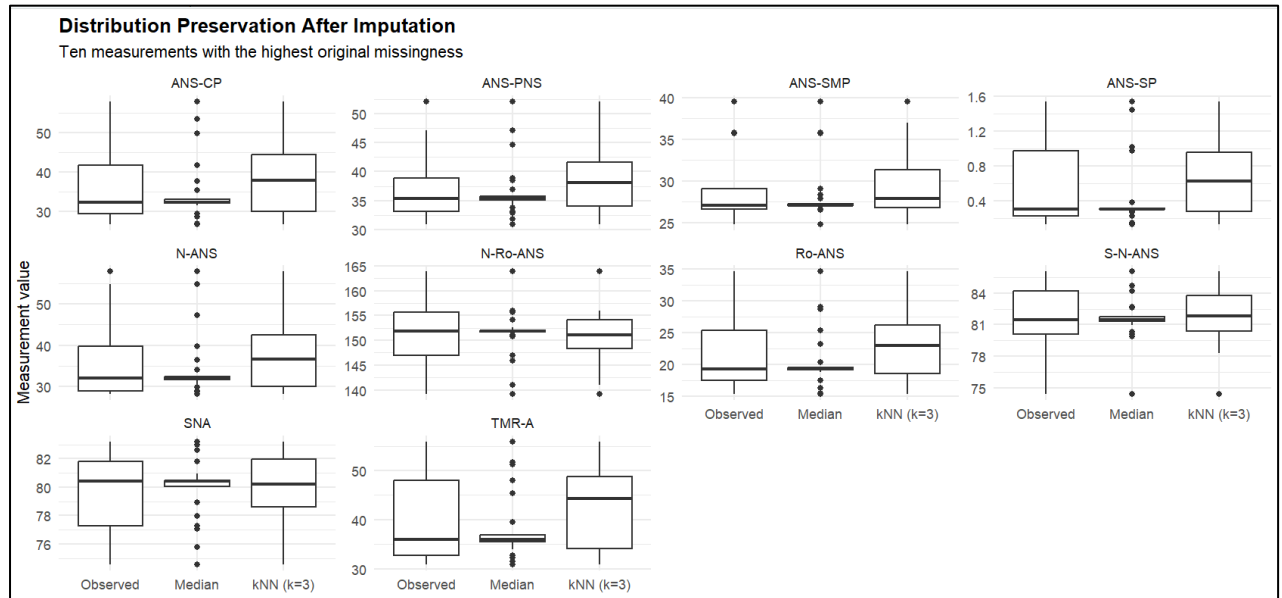


Figure 6. Distribution of observed and imputed values for the ten craniofacial measurements with the highest original missingness. Median imputation generally reduced variability, whereas kNN () preserved the observed distributions more closely.

Table 2. Comparison of kNN neighbourhood sizes

k	RMSE	MAE
3	3.8155	2.6784
4	3.9145	2.5578
5	4.0711	2.6761

Table 3. Overall validation performance by imputation method

Method	RMSE	MAE
Median	6.6835	4.9359
kNN (k=3)	3.8155	2.6784

Table 4. Validation performance by missingness level

Missingness level	N masked	Median RMSE	Median MAE	kNN RMSE	kNN MAE
Low (<10%)	56	6.25	4.68	3.43	2.32
Moderate (10–30%)	2	5.97	5.92	3.87	3.34
High (>30%)	34	7.38	5.31	4.37	3.24

Table 5. Correlation preservation after imputation

Method	Strong correlation pairs	Mean absolute change in (r)	Median absolute change in (r)
Median	4,054	0.0912	0.0106
kNN ((k=3))	4,054	0.0158	0.0032